29. Ans: A

A range of (kinetic) energies indicates a range of speeds for the β particles.
Since beta particles are emitted with a range of speeds, the products of a beta decay process cannot just consist of the daughter nuclide (product nuclide) and the beta particle as this would imply definite speeds for both products, in order for linear momentum to be conserved.

Option B is wrong. There is no such observation. Neutrino is chargeless. The total charge of the decay products is equal to the charge of the parent nuclide.

Option C is wrong. It is a true observation, but the loss in mass during β decay is due to conversion to energy released, and not the existence of the neutrino.

Option D is wrong. There is no such observation. Neutrino is chargeless so has no ionising power, and therefore cannot be observed in a cloud chamber.